AI NeuroOnco: Comprehensive Results Report

Brain Tumor MRI Classification, Clinical Benchmarks & Multimodal RAG Performance

Study Phase: Track 3 (AI Integration)

Report Date: September 17, 2026

Diagnostic Task: 4-Class Brain Tumor MRI

Modality: Axial T1/T2 & Volumetric DICOM

Primary Architecture: EfficientNet-B0 (TL)

Evaluation Cohort: 2,107 Locked Test Images

1. Datasets, Origins & Data Governance

The AI NeuroOnco research platform utilizes curated and audited neuro-imaging cohorts for model training, internal locked-set testing, and clinical retrieval grounding:

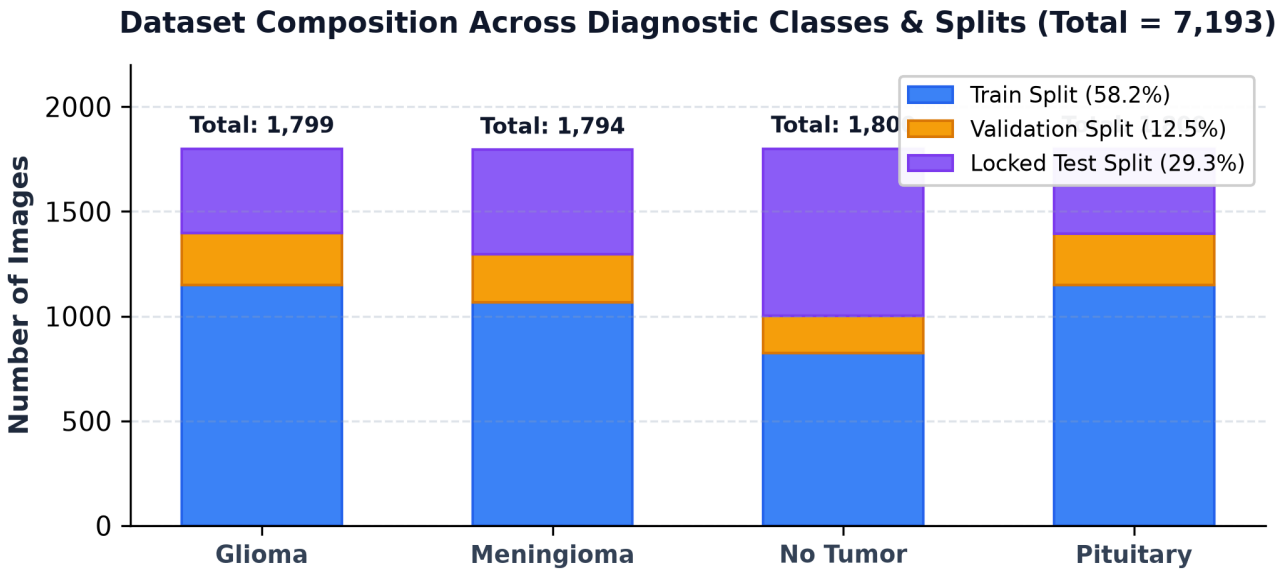
Dataset Reference	Type / Role	Sample Size	Description & Components
Brain Tumor MRI Dataset (Kaggle / masoudnickparvar V2)	Primary Development (Train / Val / Test)	**7,193** images (audited from 7,200)	Multiclass pool combining **Figshare** (Cheng et al.), **SARTAJ**, and **Br35H** datasets. CC BY 4.0 license.
BRISC 2025 (Fateh et al., arXiv:2506.14318)	Auxiliary / Localization Pixel Mask Audit	**6,000** scans (T1-weighted)	Physician-validated pixel-level tumor masks across 3 anatomical planes (axial, coronal, sagittal). Used for Grad-CAM overlap analysis.
BDNeuro-MRI v7 (Epic & CSCR Hospital)	External Candidate Deduplicated Benchmark	**5,941** scans (deduplicated)	Hospital MRI collection with exact & perceptual duplicate removal (Hamming dist <= 5). Used for baseline fine-tuning comparison.
Literature Tracker (AI_NeuroOnco RAG Corpus)	Knowledge Base Vector Store (Chroma)	**74** papers 372 text chunks	Curated full-text neuro-oncology peer-reviewed literature indexed with sentence-transformers for grounded RAG query responses.

Data Integrity & Leakage Prevention: During strict data auditing, 1 cross-label perceptual duplicate component consisting of 7 conflicting images was detected and completely excluded without visual relabelling. The finalized 7,193-image development manifest was formally audited with **zero cross-split leakage** and 100% label consistency.

2. Dataset Size & Split Breakdown

The primary dataset (7,193 images) is partitioned into Training (58.2%), Validation (12.5%), and a strictly held-out, locked Internal Test partition (29.3%). All images are preprocessed and resized to 224x224x3:

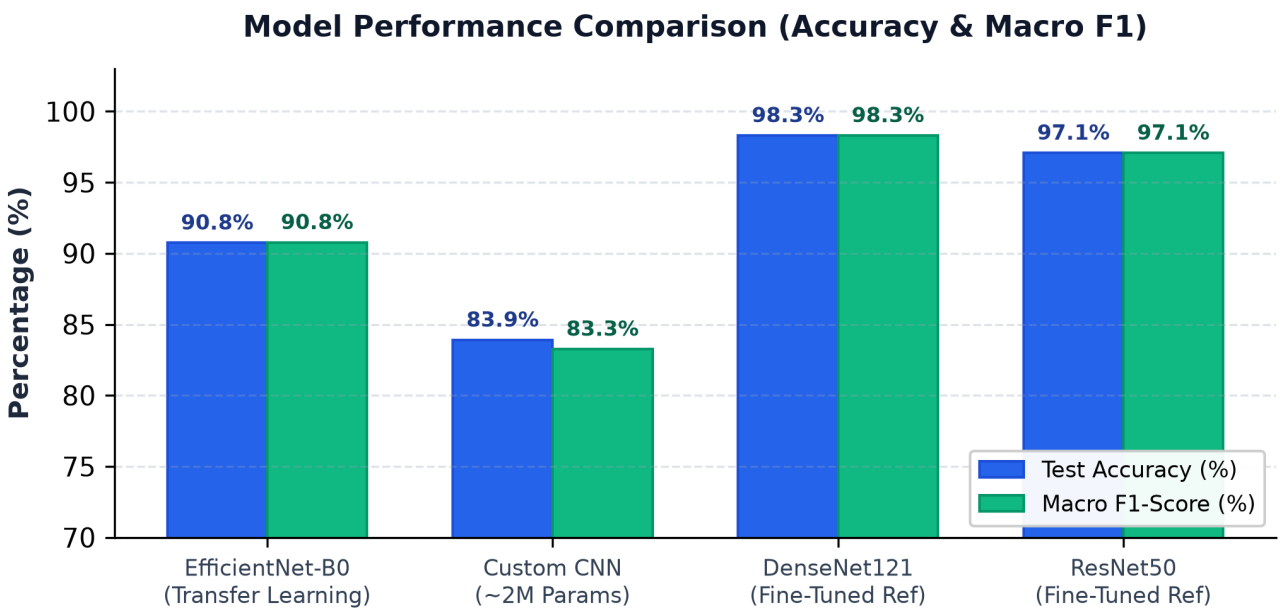
Diagnostic Class	Training Set	Validation Set	Locked Test Set	Total Cohort	Class %
Glioma	1,149	247	403	**1,799**	25.0%
Meningioma	1,066	229	499	**1,794**	24.9%
No Tumor (Healthy)	824	177	799	**1,800**	25.0%
Pituitary Tumor	1,148	246	406	**1,800**	25.0%
TOTALS	**4,187 (58.2%)**	**899 (12.5%)**	**2,107 (29.3%)**	**7,193**	**100.0%**



3. Trained Model Performance & Statistical Accuracy

Models were evaluated against the official locked internal test cohort consisting of **2,107 unseen scans** across 1,460 independent bootstrap duplicate components. Metrics report point estimates and 95% confidence intervals (CI) calculated via group-stratified bootstrapping:

Performance Metric	EfficientNet-B0 (Transfer Learning)	Custom CNN (~2M Parameters)	Statistical Significance (Paired Delta)
Overall Test Accuracy	**90.75%** (88.90% – 92.47%)	**83.91%** (81.66% – 86.11%)	+6.84% (p = 2.58e-29, McNemar)
Balanced Accuracy	**89.31%**	**81.63%**	+7.68% improvement
Macro F1-Score	**90.78%** (89.09% – 92.27%)	**83.27%** (81.24% – 85.37%)	+7.51% (95% CI: 5.7% – 9.4%)
Macro ROC-AUC (OvR)	**97.98%** (0.9798)	**95.38%** (0.9538)	+2.60% discriminative gain
Macro PR-AUC (OvR)	**96.30%** (0.9630)	**91.17%** (0.9117)	+5.13% precision-recall gain
Matthews Correlation (MCC)	**0.8759**	**0.7866**	+0.0893 correlation shift
Expected Calibration Error (ECE)	0.0525 → 0.0559 (calibrated)	0.0288 → 0.0995 (calibrated)	Temp T = 0.608 (EfficientNet)



4. Benchmark Comparisons on Reference Cohorts

To contextualize performance across convolutional and transformer architectures, standard transfer learning paradigms were benchmarked on the deduplicated BDNeuro-MRI cohort (5,941 scans):

Architecture	Training Paradigm	Test Accuracy (%)	Test Macro-F1	Parameter Complexity
DenseNet121	Fine-Tuned Pretrained	**98.31%**	**0.9831**	~7.98M parameters
ResNet50	Fine-Tuned Pretrained	**97.08%**	**0.9708**	~25.6M parameters
EfficientNet-B0	Transfer Learning (Primary)	**90.75%**	**0.9078**	~5.3M (Optimal efficiency)
DenseNet121	Frozen Pretrained Backbone	**87.06%**	0.8703	Linear classification head
Custom CNN	Trained from scratch	**84.70% / 83.91%**	0.8457 / 0.8327	~2.0M parameters (Lightweight)
Vision Transformer (ViT)	Trained from scratch	**82.56%**	0.8240	Patch projection self-attention
Hybrid CNN-ViT	Trained from scratch	**81.89%**	0.8201	Conv stem + Transformer blocks

5. Clinical AI Decision Support & Safety Frameworks

Beyond core classification accuracy, the AI NeuroOnco framework implements four critical safety and diagnostic governance mechanisms to prevent automated diagnostic errors in clinical settings:

1. Normalized Shannon Entropy Index: Calculates prediction distribution conflict: $H(P) = -\Sigma\, p_i \log2(p_i) / 2.0$. Provides quantitative tri-state decision classification: **Decisive** (< 0.300), **Ambivalent** (0.300 – 0.600), or **Conflicted** (> 0.600) requiring mandatory human radiologist review.

2. Explainable AI (Grad-CAM): Activates feature hooks on final convolutional layers to generate pixel-level attention heatmaps, enabling visual audit of tumor morphology alignment (1,045 cases quantitatively evaluated).

3. PACS / HL7 FHIR Integration: Generates standardized FHIR DiagnosticReport schemas containing LOINC/SNOMED diagnostic codes, model timestamps, prediction probability distributions, and entropy metrics for electronic health records.

4. Multimodal RAG with MedGemma: Integrates Google's medical foundation models (MedGemma / Gemini 2.0 Flash) with ChromaDB vector search across 74 indexed research papers, providing grounded literature citations and source snippet verification.

6. Research Governance & Clinical Disclaimers

Non-Diagnostic Research Notice: This software and report are intended solely for retrospective academic research and algorithm evaluation. The diagnostic labels ('glioma', 'meningioma', 'pituitary', 'no_tumor') reflect source dataset annotations and do not substitute for formal histopathological examination or WHO CNS5 molecular classification. No individual treatment or surgical decisions should be based on automated outputs without independent board-certified radiologist verification.